

# Relation Algebra

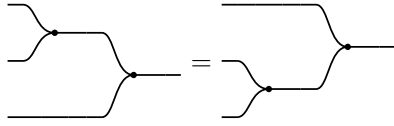
## §2. Relations

(2a)

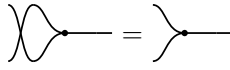
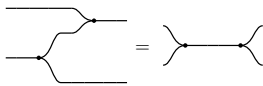
$\mathbf{Rel}$  is a poset-enriched category with  $(\otimes, \circ)$  where  $\otimes$  is commutative cartesian product, and converse  $\circ \dashv \circ$ , and

$\mathbf{C}: (\triangleright : A \otimes A \rightarrow A, \circ : I \rightarrow A)$  a commutative monoid with  $\mathbf{C} \circ : \mathbf{C}$ . We write  $\mathbf{C} \circ$  as  $(\triangleleft : A \rightarrow A \otimes A, \circ : A \rightarrow I)$

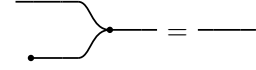
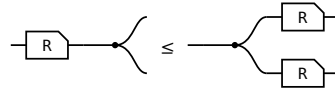
(2b)



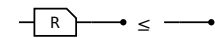
$\triangleright$  associative



$\triangleright$  commutative



$\circ$  is its unit



Frobenius, one half — the other is its  $\circ$

$R \triangleleft \leq \triangleleft (R \otimes R)$

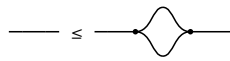
$R \circ \leq \circ$

### §2.1. $\forall$ object $A$ , $(A, \triangleleft, \circ) \dashv (A, \triangleright, \circ)$

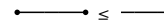
(2.1a)



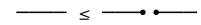
$\triangleleft \leq 1$



$1 \leq \triangleleft$



$\circ \leq 1$



$1 \leq \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In  $\mathbf{Rel}$  it reads  $\{(a, a)\} \subseteq A \times A$  — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.